11. In Ms Kirkpatrick's English class there are 12 girls and 18 boys. She wants to select the greatest number of debating teams from her class so that (i) no student is left out (ii) the same number of girls / boys is on each team. How many teams can she make and what is the composition of each team?

12. Write 26 as the sum of two primes in three different ways.

13. Which of these numbers are prime?
(i) 7 (ii) 19 (iii) 26 (iv) 31 (v) 39

14. Continue the sequence 6, 12, 18, … as far as 48.
Now write down the numbers which are one smaller and one larger than this sequence, i.e., 5, 7, 11, … 49.
Which of these numbers are **not** prime?

15. John has 24 chocolate bars and 20 packets of crisps. How many friends could he share them with so that he has none left over and each friend gets the same number of each?

16. Are the following statements true or false? If false, say why.
(i) All prime numbers are odd numbers.
(ii) The only even prime number is 2.
(iii) All odd numbers are prime numbers.
(iv) There are only three prime numbers between 10 and 20.

17. Write down all the factors of 48.
How many of these factors are prime?

18. Peter scored 56 points and Sarah 48 points having played the same number of computer games. What was the greatest number of games each could have played? Find also the number of points per game each was awarded. (Note, each game is awarded the same number of points)

19. Write down the prime factors of each of these numbers:
(i) 12 (ii) 30 (iii) 45 (iv) 84 (v) 108

20. Find a pair of prime numbers that total each of these numbers:
(i) 12 (ii) 16 (iii) 24 (iv) 32

21. **Twin primes** are prime numbers with a difference of 2. The first pair of twin primes is 3 and 5. What are the next three pairs of twin primes?

22. Think of a number between 1 and 13.
Multiply it by one less than the number you have chosen and then add 41.
Do this a few times. Comment on your answers.

23. Barry has 16 white, 40 green and 24 orange beads. How many bracelets can he make using all of the beads if he wants to have the same number of each colour on each bracelet.
How many beads of each colour are on each bracelet?

24. Look at the numbers 3, 4, 5, 9, 10, 12, 15, 18, 30.
(i) Which numbers are prime?
(ii) Which numbers are factors of 30?
(iii) Which numbers have 6 as a factor?
(iv) How many of the numbers are divisible by 5?

Section 1.3 Multiples – Lowest common multiple

If we count in groups of 3 we produce multiples of 3.

1 2 **3** 4 5 **6** 7 8 **9** 10 11 **12** 13 14 **15** 16 17 **18** 19 20 **21**

The multiples of 3 are 3, 6, 9, 12, 15, 18, **21**, 24, …

Counting in groups of 7 we produce multiples of 7.

1 2 3 4 5 6 **7** 8 9 10 11 12 13 **14** 15 16 17 18 19 20 **21** …

The multiples of 7 are 7, 14, **21**, 28, 35, …

Notice that **21** is the smallest number common to both sets of multiples.

21 is called the **lowest common multiple (LCM)** of 3 and 7.

Lowest Common Multiple

The lowest common multiple (LCM) of two or more numbers is the smallest number into which each of the numbers divide.

Example 1

Find the lowest common multiple of 8, 12, 18.

The multiples of 8 are: 8, 16, 24, 32, 40, 48, 56, 64, **72**, …
The multiples of 12 are: 12, 24, 36, 48, 60, **72**, 84, …
The multiples of 18 are: 18, 36, 54, **72**, …
The lowest common multiple (LCM) = 72.